PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 09/ 10)
Reporting Period: From 10/08/2026 To 17/09/2026

1. Student Information

Student Name	PATEL MAN VISHNUBHAI.
Enrollment Number	25IC03CA053
Program/Branch	MCA (CA)/ ICA
Semester	3
Contact Number	8849948963
Email ID	25ic03ca053@ppsu.ac.in

2. Project Details

Type	Research Project
Department/Domain	MCA CA (DS/ML) /AI &ML
Mentor Name	Mr. Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Complete ZIP Download Functionality	Completed	ZIP Download feature completed and integrated into the project.
Final System Testing and Debugging	Completed	Tested the complete system and resolved identified errors.
Validate Complete Project Workflow	Completed	Verified the workflow from image upload to final report generation.
Fix Remaining Errors and Issues	Completed	Resolved major issues and improved system reliability.
Finalize Project Documentation	Completed	Prepared project documentation, screenshots, and result evidence.
Prepare Final Project Demonstration	Completed	Completed preparation of the final project presentation and demonstration.

B. Work Completed During This Reporting Period

Description:

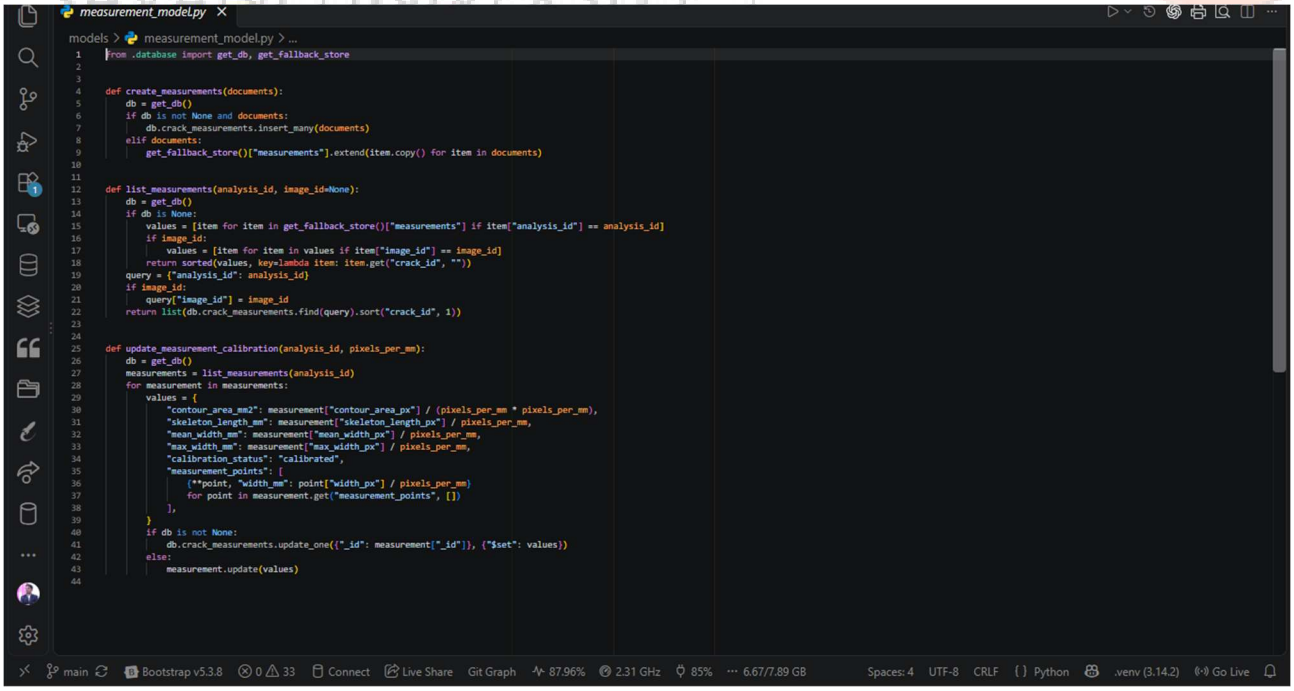
- Completed the remaining development tasks and finalized the complete project workflow.
- Completed ZIP Download functionality and integrated all project modules.
- Finalized crack detection, width estimation, risk classification, result visualization, and CSV reporting.
- Reviewed relevant research papers and finalized project references.
- Performed final system testing and debugging using multiple crack images.
- Verified the complete workflow and resolved identified issues.
- Successfully generated annotated images, CSV reports, and downloadable ZIP packages.
- Completed final project documentation, screenshots, and demonstration preparation.

C. Evidence / Deliverables Produced

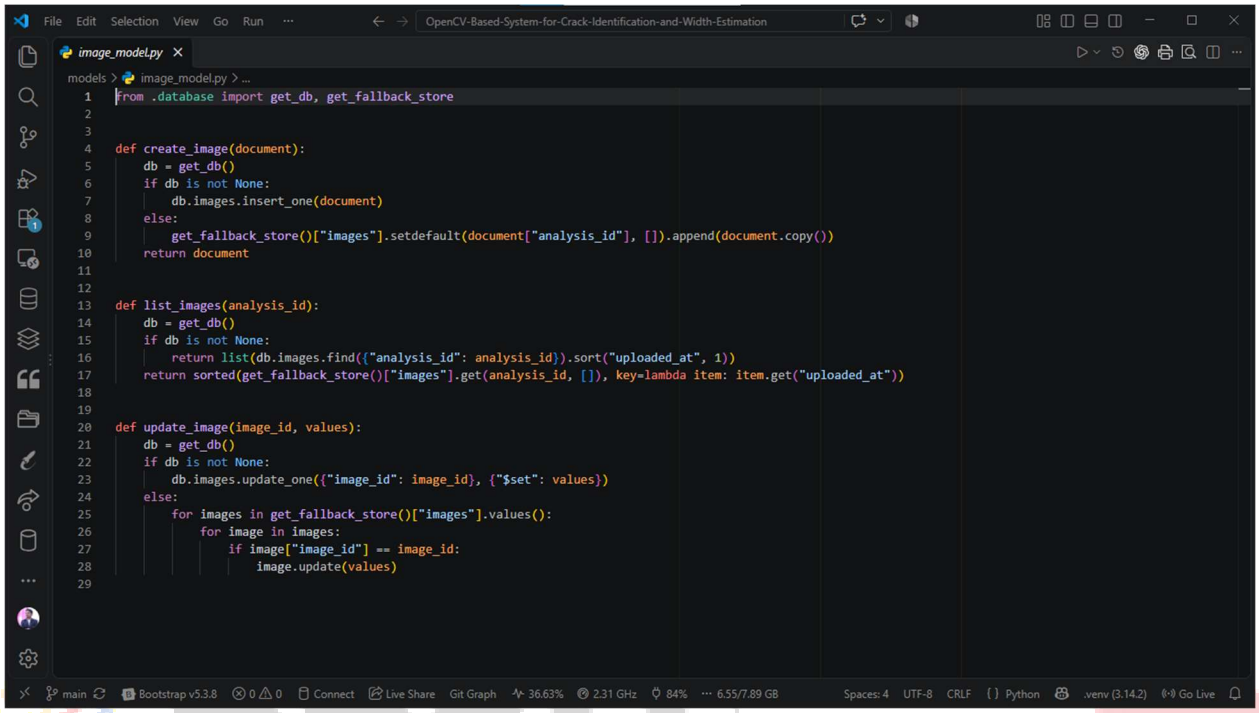
Sr. No.	Deliverable / Evidence	Description / Link / Attachment
1	Source Code	Completed Python source code with all integrated crack detection and measurement modules.
2	Screenshots and Model Output	Annotated crack images showing detection, width measurements, and risk classification results.
3	CSV and ZIP Reports	Generated CSV analysis report and completed ZIP package containing project outputs.
4	Test Results	Final testing and validation results using multiple crack images.
5	Research Paper Review	Reviewed research papers related to crack detection, image processing, and crack-width estimation.
6	SRS / Design Document	System workflow, project methodology, and technical design details.
7	Project Presentation	Project demonstration materials and presentation preparation.
8.	Project Repository	GitHub: https://github.com/ManPatel448/OpenCV-Based-System-for-Crack-Identification-and-Width-Estimation.git

- **Screenshots**

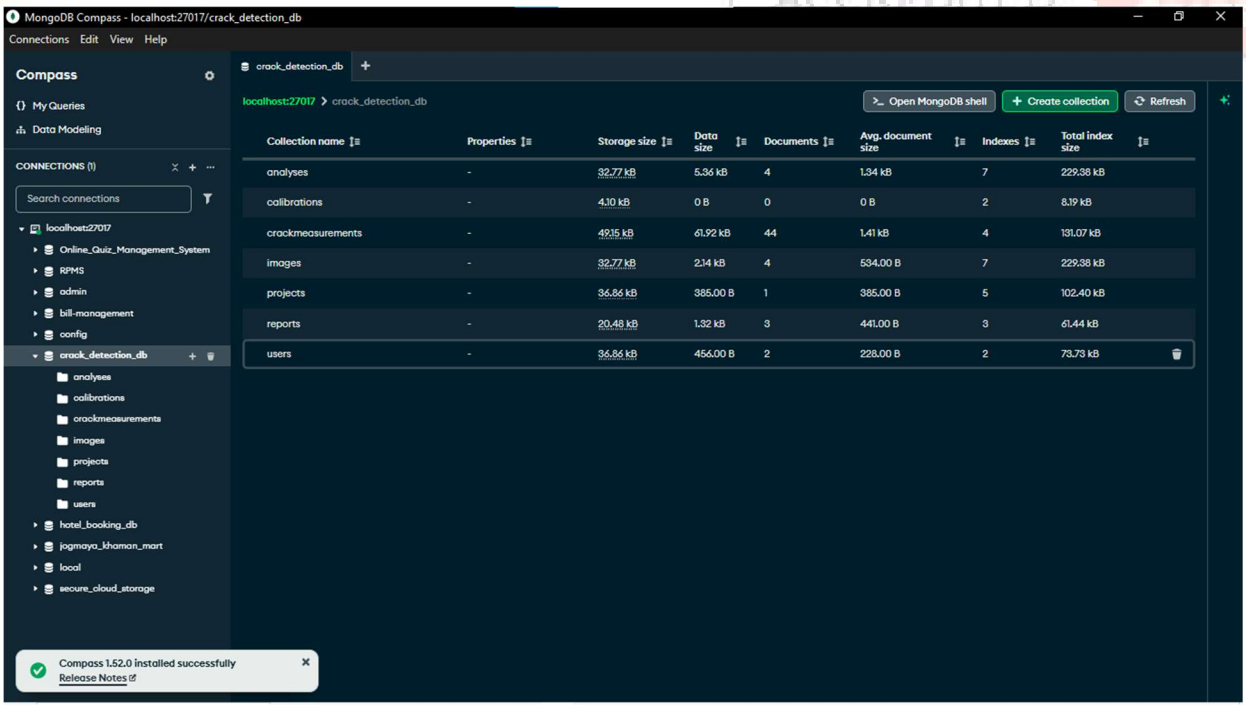
measurement_model.py



Image_model.py



MongoDB Database:



D. Current Project Progress

Overall Project Completion: 97%

Sr. No.	Project Component / Module	Status	Completion (%)
1	Library Installation	Completed	100%
2	Import Libraries	Completed	100%
3	Project Configuration	Completed	100%
4	Image Upload	Completed	100%
5	Image Preprocessing	Completed	100%
6	Crack Segmentation	Completed	100%
7	Crack Detection	Completed	100%
8	Crack Width Estimation	Completed	100%
9	Risk Classification	Completed	100%
10	Result Visualization	Completed	100%
11	CSV Report	Completed	100%
12	ZIP Download	Completed	100%

E. Challenges / Issues Faced

- Faced minor issues during final system integration and testing.
- Encountered challenges in ensuring accurate crack detection under different lighting and image conditions.
- Required parameter adjustments to improve crack-width estimation and measurement consistency.
- Faced file-handling challenges while generating and downloading ZIP packages.
- Resolved errors related to output generation and verified the complete project workflow.
- Completed final testing and improved the overall system reliability.

F. Actions Taken / Solution Implemented

- Optimized OpenCV parameters to improve crack detection and image-processing results.
- Applied pre-processing and morphological operations to reduce noise and improve crack segmentation.
- Verified pixel-to-mm calibration to improve crack-width measurement consistency.
- Implemented debugging and error handling to resolve system issues.
- Improved ZIP file generation and organized output files for easy downloading.
- Performed final testing using multiple crack images and validated all modules.
- Integrated all modules into a complete and reliable project workflow.

G. Plan for Next Reporting

- Prepare the main project documentation with complete system details, methodology, and results.
- Complete the research paper writing based on the project methodology and findings.
- Review and organize research papers, references, and citations.
- Identify and fix minor project bugs to improve system stability.
- Perform final testing and verify all project modules.
- Finalize project screenshots, result analysis, and documentation for submission.
- Prepare the final project presentation and demonstration.

H. Learning / Skills Acquired

- Improved Python programming and OpenCV image-processing skills.
- Gained practical experience in crack detection, segmentation, and width estimation.
- Learned advanced techniques for image calibration, risk classification, and result analysis.
- Improved debugging, testing, and minor bug-fixing skills.
- Developed skills in technical documentation and research paper writing.
- Improved literature review, reference management, and research analysis.
- Gained experience in project maintenance, system integration, and final validation.
- Strengthened technical writing, presentation, and co mmunication skills.

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5. Tasks Accomplished

- Completed all major modules of the crack identification and width estimation system.
- Successfully implemented and integrated the ZIP Download functionality.
- Completed final system testing and validation using multiple crack images.
- Verified crack detection, width estimation, and risk classification results.
- Generated annotated images, CSV reports, and downloadable ZIP packages.
- Resolved major integration issues and improved system reliability.
- Organized project outputs and finalized the system workflow.
- Completed the main development phase and started preparing project documentation and research paper work.

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6. Tasks in Progress / Next Plan

- Complete the main project documentation with system design, methodology, and results.
- Continue research paper writing based on project implementation and findings.
- Organize research papers, references, and citations.
- Identify and fix minor bugs in the project.
- Perform additional testing to improve system stability and reliability.
- Finalize project screenshots, result analysis, and documentation.
- Prepare the final project presentation and demonstration.

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7. Learnings / Skills Acquired

- Improved Python programming and OpenCV image-processing skills.
- Gained knowledge of crack detection, segmentation, and width measurement.
- Developed skills in debugging, testing, and minor bug fixing.
- Improved technical documentation and research paper writing skills.
- Learned effective research analysis and reference management techniques.
- Strengthened problem-solving, project maintenance, and system validation skills.
- Improved technical communication and project presentation skills.

References:

- Bodke, K., Bhirud, S., & Sangle, K. K. (2026). Enhanced Crack Width and Depth Measurement through Binary Image Processing and Geometric Analysis. Scientific Visualization, 18(1), 49–61.

- Ramasamy, L., & Vijayan, P. (2026). Advanced Image Processing and Deep Learning Methods for Automated Crack Detection in Civil Engineering Structures: A Comprehensive Review. Asian Journal of Civil Engineering.
- Deng, J., Singh, A., Zhou, Y., Lu, Y., & Lee, V. C.-S. (2022). Review on Computer Vision-Based Crack Detection and Quantification Methodologies for Civil Structures. Construction and Building Materials, 356, 129238.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3– Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4– Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5– Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4– Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	
Planning for Next Phase	L6– Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3– Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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